LM555 TIMER CIRCUIT GROUP PROJECT

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COURSE: EED 1009

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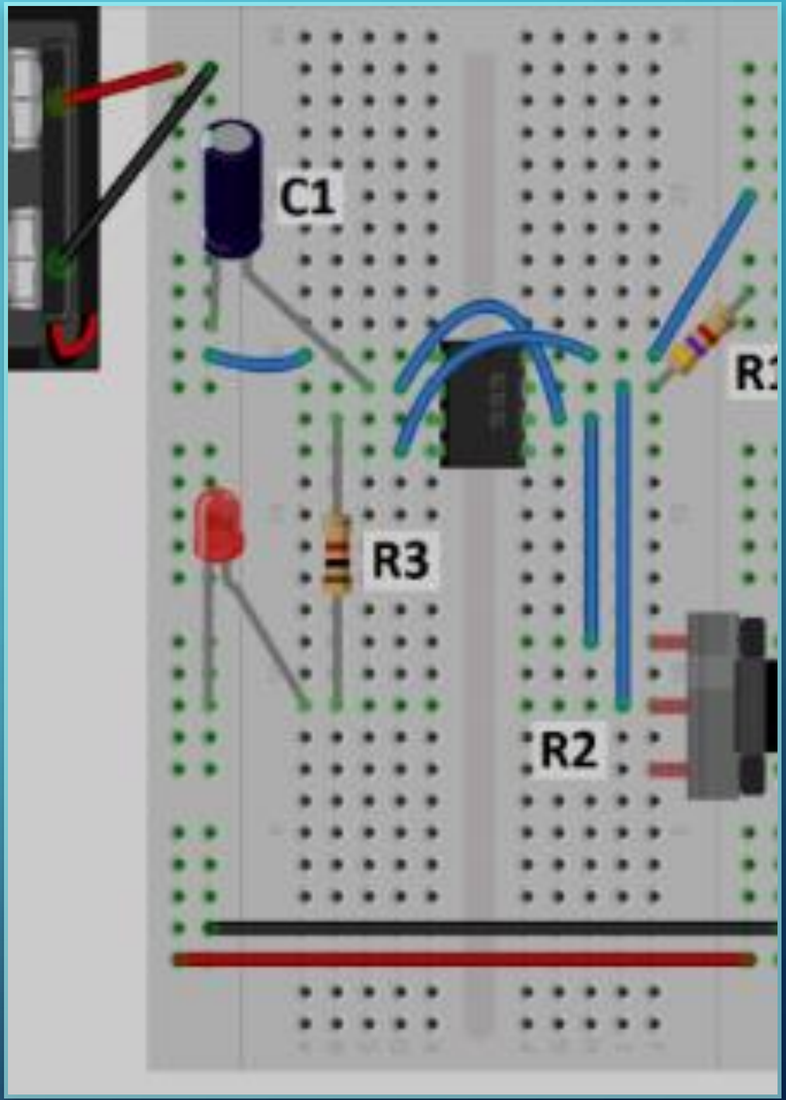
WHAT WE'LL BE GOING THROUGH IN THIS PRESENTATION:

*Information about LM555 and other components

- Information about the design of our circuit
- How we chose the circuit elements for this project
- Values of the circuit elements we've used
- Problems we've faced
- Budget analysis of the circuit
- Conclusion
- References



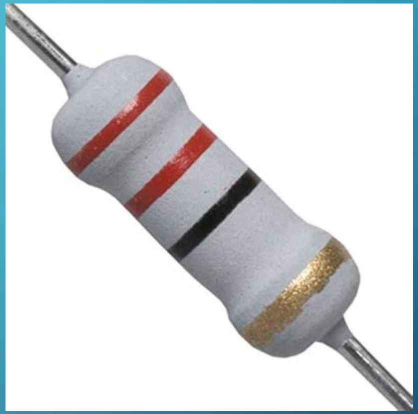
INFORMATION ABOUT LM555

- **LM555 : Widely used in Engineering and very versatile.**

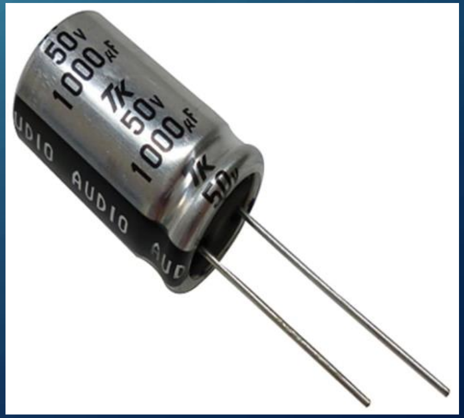
- **Modes:** Astable, Monostable, Bistable

- **Application: Used in timing and wave generation**

INFORMATION ABOUT OTHER COMPONENTS

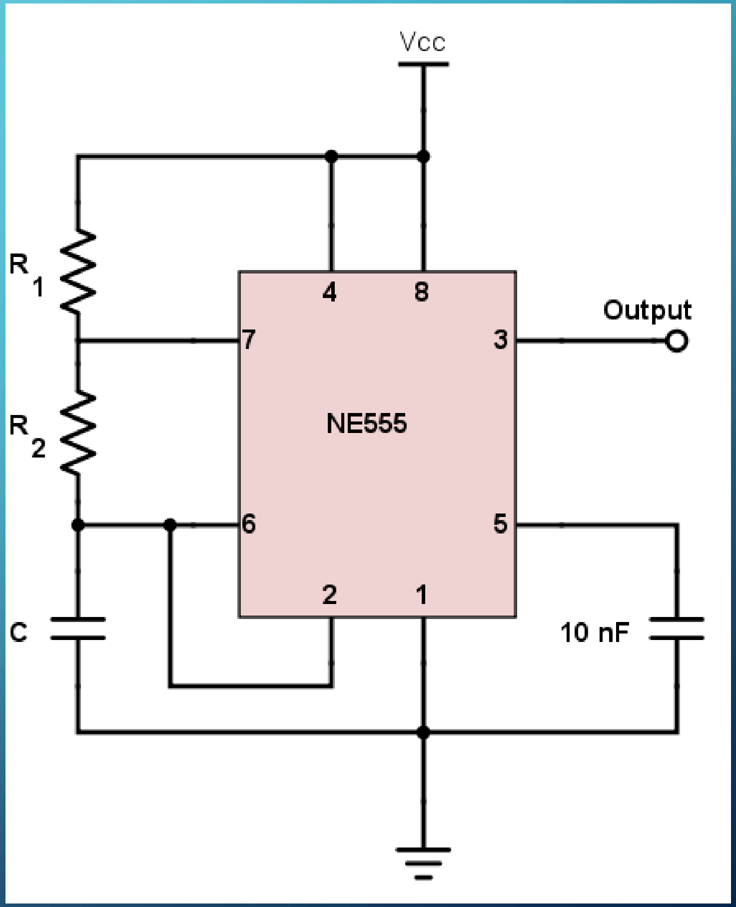


- Resistors and capacitors: Key components for tuning circuit parameters.
- Potentiometer: Adjusts frequency in astable mode.
- Battery: Provides voltage to the circuit
- LED : Outputs light



INFORMATION ABOUT THE DESIGN OF THE CIRCUIT

- LM555 configured in astable mode for continuous output.
- Generates a square wave to control LED blinking.
- A simple, functional design for visualizing frequency changes.
- The circuit diagram is shown for a better understanding.



CHOOSING CIRCUIT ELEMENTS

- LM555 IC: Easy to use and widely available.
- Resistors and capacitor: Calculated for the desired frequency.
- Potentiometer: Allows real-time frequency adjustment.
- LED: Chosen to visually display the output signal.
- Power source: Selected to match LM555's voltage range

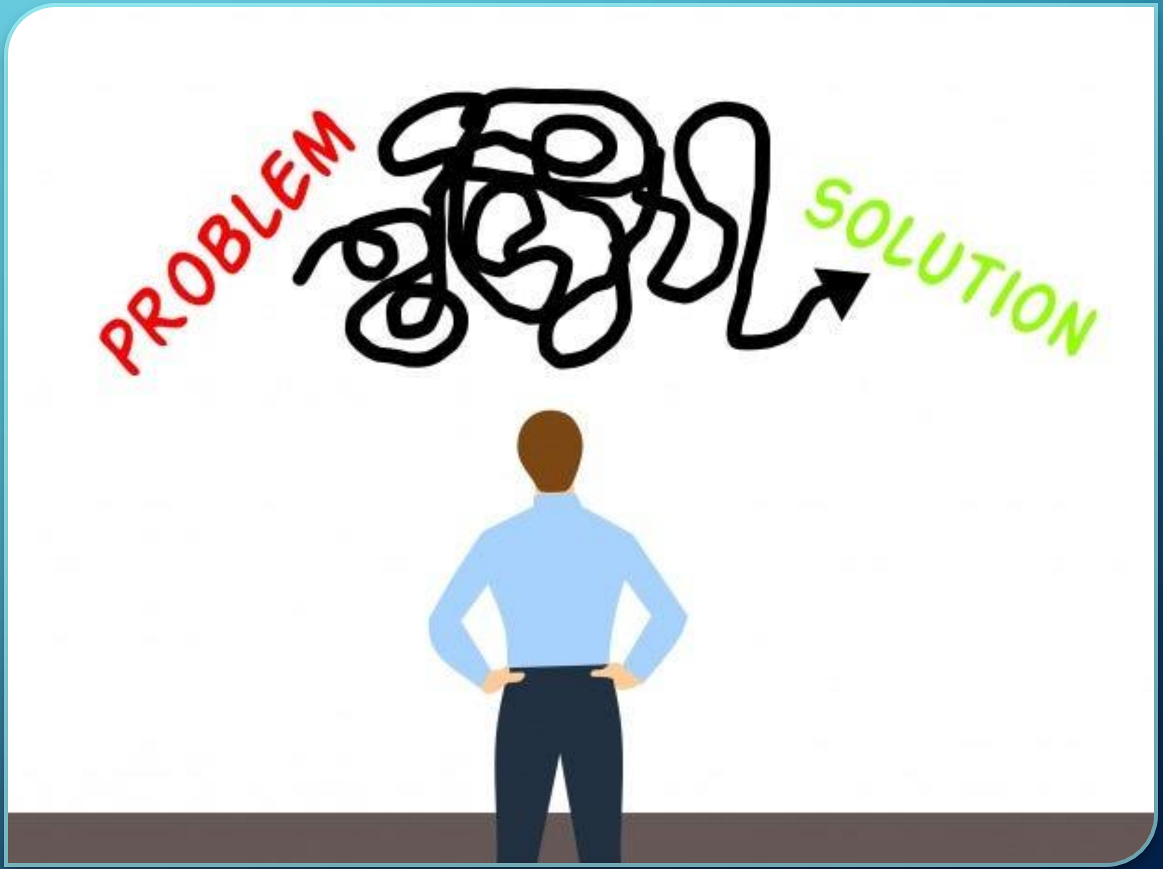
CIRCUIT ELEMENTS VALUES

$$T = \frac{1}{f} = 0.694(R_1 + 2R_2)C$$

- Resistors: 1.6kΩ
- Capacitor: 25 V 47 µF
- Potentiometer: 10kΩ (adjustable)
- Power supply: 9V DC

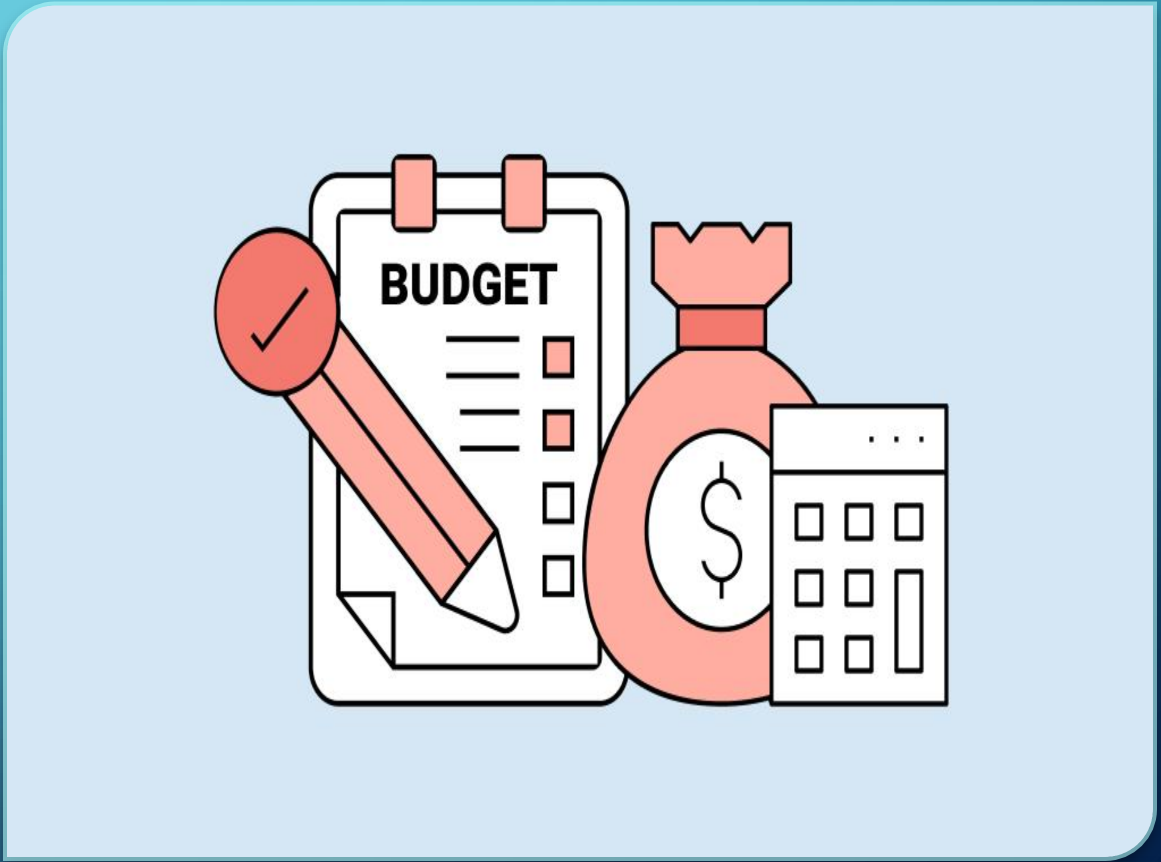
PROBLEMS WE'VE FACED

- Being unable to light the LED lamp
  Camera not being able to capture the change in period
- LED lamp exploding due to high current



BUDGET

- Total cost: 120₺
- Reusable components: LM555 IC, breadboard, wires.
- Consumables: Resistors, capacitors, and LED.
- Economical design with potential for reuse.



CONCLUSION

- The LM555 IC helped us create a
- Variable - period LED circuit.

- We learned how to design and troubleshoot circuits.

- This project showed how theory can be applied in practice.

THANK YOU FOR LISTENING!

REFERENCES

- 555 Timer Astable Circuit," AllAboutCircuits.com
- 555 Timer Basics," CircuitBasics.com